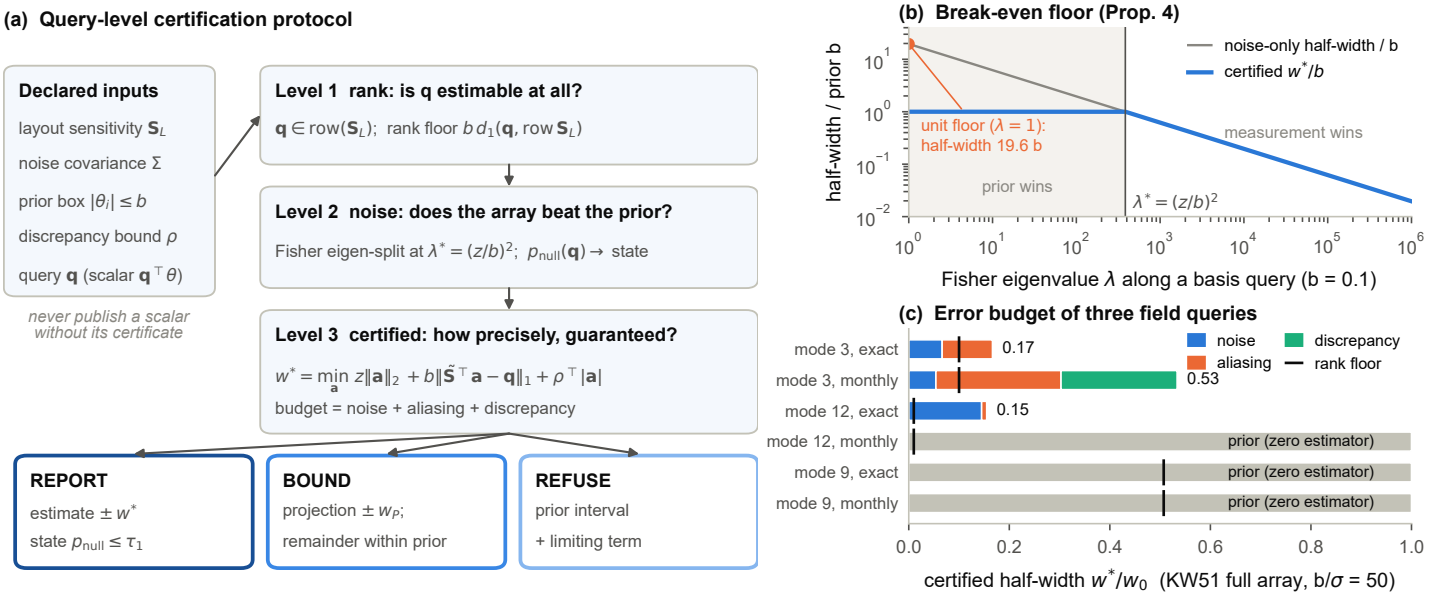

(a) Query-level certification protocol
Declared inputs
layout sensitivity $\mathbf{S}_L$
noise covariance Σ
prior box $|\theta_i| \le b$
discrepancy bound ρ
query $\mathbf{q}$ (scalar $\mathbf{q}^\top\theta$)
never publish a scalar without its certificate
Level 1 rank: is q estimable at all?
$\mathbf{q} \in \mathrm{row}(\mathbf{S}_L)$; rank floor $b\, d_1(\mathbf{q}, \mathrm{row}\,\mathbf{S}_L)$
Level 2 noise: does the array beat the prior?
Fisher eigen-split at $\lambda^* = (z/b)^2$; $p_{\mathrm{null}}(\mathbf{q}) \to$ state
Level 3 certified: how precisely, guaranteed?
$w^* = \min_{\mathbf{a}} z\|\mathbf{a}\|_2 + b\|\tilde{\mathbf{S}}^\top\mathbf{a} - \mathbf{q}\|_1 + \rho^\top|\mathbf{a}|$
budget = noise + aliasing + discrepancy
REPORT
estimate $\pm\, w^*$
state $p_{\mathrm{null}} \le \tau_1$
BOUND
projection $\pm\, w_P$;
remainder within prior
REFUSE
prior interval
+ limiting term
(b) Break-even floor (Prop. 4)
half-width / prior b
noise-only half-width / b
certified w^*/b
unit floor ($\lambda = 1$): half-width 19.6 b
measurement wins
prior wins
$\lambda^* = (z/b)^2$
Fisher eigenvalue λ along a basis query (b = 0.1)
(c) Error budget of three field queries
noise
aliasing
discrepancy
rank floor
mode 3, exact
mode 3, monthly
mode 12, exact
mode 12, monthly
mode 9, exact
mode 9, monthly
0.17
0.53
0.15
prior (zero estimator)
certified half-width w^*/w_0 (KW51 full array, b/σ = 50)